低速无人车线控底盘生产下线检测报告

1. 检测对象与结论

会话 ID	EOL-1E0EDEEC82C1	检测结论	PASS
底盘编号	PH3-NORMAL-001	VIN	PH3NORMAL0000001
序列号	PH3-1783678172371	工位	EOL-STATION-01
操作员	dev-operator	检测方案	default_chassis_eol_v1
开始时间	2026-07-10T10:09:32.387127+00:00	结束时间	-
失败原因	-	安全停车	-

2. 追溯信息

software_version	v1.0.2
dbc_hash	387ae48bd84852c8
config_hash	profile:dev
config_profile	dev
test_plan_id	default_chassis_eol_v1
test_plan_version	2.0.0
operator	dev-operator

3. 检测步骤

序号	步骤	状态	结果	耗时(ms)	失败原因
1	上电自检	DONE	PASS	105	-
2	CAN 通信检测	DONE	PASS	211	-
3	BMS 检测	DONE	PASS	256	-
4	VCU 状态检测	DONE	PASS	258	-
5	电机心跳检测	DONE	PASS	207	-
6	档位检测	DONE	PASS	1610	-
7	低速驱动检测	DONE	PASS	2484	-
8	转向检测	DONE	PASS	4031	-
9	灯光检测	DONE	PASS	4435	-
10	制动停止检测	DONE	PASS	2591	-
11	告警复查	DONE	PASS	201	-
12	报告生成	DONE	PASS	-	-

4. 检测断言

步骤	断言	信号	阈值	测量值	来源	结果
上电自检	数据库可写	system	true	true	good/	PASS
上电自检	DBC 已加载	system	true	true	good/	PASS
上电自检	急停已释放	system	true	true	good/	PASS
CAN 通信检测	CAN1 原始接收在线	system	true	true	good/	PASS
CAN 通信检测	CAN2 原始接收在线	system	true	true	good/	PASS
CAN 通信检测	关键报文完整	0x51,0x77,0x168,0xE1	["0x168", "0x51", "0x77", "0xE1"]	{"present": ["0x168", "0x51", "0x77", "0xE1"], "missing": []}	good/	PASS
BMS 检测	BMS 总压在范围内	BMS_Voltage	{"min": 36.0, "max": 72.0, "note": "site_confirm_battery_pack_voltage"}	52.0	good/0x100	PASS
BMS 检测	BMS SOC 不低于阈值	BMS_SOC	30	86	good/0x100	PASS
BMS 检测	NTC 温度均在范围内	NTC1,NTC2,NTC3	{"min": -20, "max": 60}	[25, 26, 24]	good/0x103	PASS
BMS 检测	0x102 保护位均未触发	0x102	0	{"BMS_Protect_Bit_00": 0, "BMS_Protect_Bit_01": 0, "BMS_Protect_Bit_02": 0, "BMS_Protect_Bit_03": 0, "BMS_Protect_Bit_04": 0, "BMS_Protect_Bit_05": 0, "BMS_Protect_Bit_06": 0, "BMS_Protect_Bit_07": 0, "BMS_Protect_Bit_08": 0, "BMS_Protect_Bit_09": 0, "BMS_Protect_Bit_10": 0, "BMS_Protect_Bit_11": 0, "BMS_Protect_Bit_12": 0, "BMS_Protect_Bit_13": 0, "BMS_Protect_Bit_14": 0, "BMS_Protect_Bit_15": 0, "BMS_Protect_Short_Circuit": 0, "BMS_Protect_Over_Current": 0, "BMS_Protect_Ove… [已省略 204 字符，完整值见 JSON/CSV 报告]	good/0x102	PASS
VCU 状态检测	点火状态有效	CCU_Ignition_Status	null	1	good/0x51	PASS
VCU 状态检测	初始车速接近零	CCU_Vehicle_Speed	0.2	0.0	good/0x51	PASS

步骤	断言	信号	阈值	测量值	来源	结果
VCU 状态检测	无非预期制动冲突	Remote_Brake_Request_Status,Touch_Brake_Request_Status	null	[false, false]	good/0x51	PASS
电机心跳检测	双电机心跳在线	0x703,0x704	["0x703", "0x704"]	{"present": ["0x703", "0x704"], "missing": []}	good/	PASS
档位检测	档位反馈跟随命令	CCU_Shift_Level_Status	"feedback follows D/N/R/N"	[{"expected": 0, "actual": 0, "observed": [0.0, 0.0, 0.0, 0.0, 0.0], "matched": true}, {"expected": 1, "actual": 1, "observed": [1.0, 1.0, 1.0, 1.0, 1.0], "matched": true}, {"expected": 3, "actual": 3, "observed": [3.0, 3.0, 3.0, 3.0, 3.0], "matched": true}, {"expected": 0, "actual": 0, "observed": [0.0, 0.0, 0.0, 0.0, 0.0], "matched": true}]	good/	PASS
低速驱动检测	车速达到目标附近	CCU_Vehicle_Speed	{"target": 1.5, "tolerance": 0.5}	[0.1, 0.3, 0.4, 0.6, 0.7, 0.9, 1.0, 1.2, 1.3, 1.4, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5]	good/0x51	PASS
低速驱动检测	四轮轮速一致	Wheel_Speed_Front_Left_RPM,Wheel_Speed_Front_Right_RPM,Wheel_Speed_Rear_Left_RPM,Wheel_Speed_Rear_Right_RPM	{"max_delta": 30.0}	[[200.0, 200.0, 200.0, 200.0]]	good/0x168	PASS
转向检测	前转向反馈跟随命令	SAS_Front_Angle	{"max_error": 8.0}	[{"expected": 30.0, "actual": 29.9, "error": 0.10000000000000142, "sample_count": 20}, {"expected": -30.0, "actual": -29.9, "error": 0.10000000000000142, "sample_count": 20}, {"expected": 0.0, "actual": 0.0, "error": 0.0, "sample_count": 20}]	good/0xE1	PASS

步骤	断言	信号	阈值	测量值	来源	结果
转向检测	后转向反馈跟随命令	SAS_Rear_Angle	{"max_error": 8.0}	[{"expected": 30.0, "actual": 29.9, "error": 0.10000000000000142, "sample_count": 20}, {"expected": -30.0, "actual": -29.9, "error": 0.10000000000000142, "sample_count": 20}, {"expected": 0.0, "actual": 0.0, "error": 0.0, "sample_count": 20}]	good/0xE1	PASS
灯光检测	灯光反馈跟随命令	Left_Turn_Light_Status,Right_Turn_Light_Status,Position_Light_Status,Low_Beam_Status	"feedback follows light commands"	[{"signal": "Left_Turn_Light_Status", "expected": true, "actual": true, "matched": true}, {"signal": "Right_Turn_Light_Status", "expected": true, "actual": true, "matched": true}, {"signal": "Position_Light_Status", "expected": true, "actual": true, "matched": true}, {"signal": "Low_Beam_Status", "expected": true, "actual": true, "matched": true}]	good/	PASS
制动停止检测	制动后车速归零	CCU_Vehicle_Speed	0.2	0.0	good/0x51	PASS
制动停止检测	制动反馈已激活	Brake_Status	true	[true, true, true, true, true, true, true, true, true, true, true, true, true, true, true, true, true, true, true]	good/0x51	PASS
告警复查	最高告警等级为零	0x77	0	0	good/0x77	PASS
告警复查	无关键通道超时	can_statistics	"all channels healthy"	{"unhealthy_channels": []}	good/	PASS

注：完整值见 JSON/CSV 报告。

5. 结论与签名

检测结论：PASS；通过步骤 12/12；失败断言 0/24。

操作员签名：________________ 质量工程师：________________